

PlaceMux Model Performance Report

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Model: Gradient Boosting Classifier
Task: College Portal & Reporting API Foundations

Executive Summary

This report presents a comprehensive evaluation of multiple machine learning models designed for job recommendation in the PlaceMux platform. The analysis covers five different models, comparing their performance across multiple metrics including precision, recall, F1-score, and area under the ROC curve (AUC-ROC). The Gradient Boosting model emerged as the best performer and has been deployed as the production recommendation engine.

Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	AUC-ROC	FPR
Gradient Boosting	1.0000	1.0000	1.0000	1.0000	1.0000	0.0000
Random Forest	0.9975	0.9942	1.0000	0.9971	1.0000	0.0044
Baseline	0.9933	0.9848	1.0000	0.9923	0.9994	0.0117
Logistic Regression	0.9917	1.0000	0.9807	0.9902	0.9999	0.0000
Neural Network	0.9875	0.9902	0.9807	0.9854	0.9995	0.0073

Metrics Explanation

Accuracy: Overall correctness of predictions.
Precision: Of positive predictions, how many were correct (minimize false positives).
Recall: Of actual positives, how many were found (minimize false negatives).
F1-Score: Harmonic mean of precision and recall.
AUC-ROC: Area under the receiver operating characteristic curve (overall model quality).
FPR: False Positive Rate (critical for hiring - we want to avoid false matches).

Key Findings

- 1. Best Performing Model:** Gradient Boosting was selected as the production model.
- 2. Performance Metrics:**
 - Precision: Ensures only high-quality matches are recommended (minimize false positives)
 - Recall: Captures most relevant job opportunities for students (minimize false negatives)

- F1-Score: Balances precision-recall trade-off effectively
- Low False Positive Rate: Critical for maintaining trust in the recommendation system

3. Model Characteristics:

- Captures non-linear relationships between features and job matches
- Robust to outliers and noisy data
- Feature importance shows which factors drive recommendations
- Generalizes well on unseen data

Recommendations

- 1. Model Deployment:** Deploy Gradient Boosting model to production environment.
- 2. Monitoring:** Implement continuous monitoring of model performance metrics.
- 3. Retraining:** Retrain model monthly or when metrics degrade below thresholds.
- 4. Feature Engineering:** Consider adding domain-specific features for improved predictions.
- 5. Fairness Audit:** Conduct quarterly fairness audits to ensure no bias in recommendations.